Personal Reading Report: ViT-DD: Multi-Task Vision Transformer for Semi-Supervised Driver Distraction Detection

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- **Role:** Frame-Level Spatial Feature Classification & Vision Transformer Implementation
- **Paper Assessed:** ViT-DD: Multi-Task Vision Transformer for Semi-Supervised Driver Distraction Detection (Ma & Wang, NeurIPS 2022 ML4AD Workshop)

1. Overview & Problem Addressed

Distracted driving is a primary catalyst for highway collisions, contributing to thousands of fatal roadway crashes annually. This issue is further amplified in SAE Level 3 conditional autonomous vehicles: when automated driving features are engaged, drivers frequently over-trust the system, disengage their attention, and exhibit delayed reaction times when emergency manual takeovers are requested. Robust in-cabin Driver Monitoring Systems (DMS) are therefore essential for rapid hazard prevention.

Historically, in-cabin visual posture detection has been dominated by 2D Convolutional Neural Networks (CNNs). However, standard CNNs rely on localized convolutional receptive fields, making it difficult to capture long-range spatial correlations across the cabin (such as correlating driver eye gaze, face angle, and hand-wheel placement). Furthermore, training auxiliary facial behavior models typically requires dense facial annotations, which are absent in standard distracted driving benchmarks.

The ViT-DD paper addresses these challenges by introducing the first pure Vision Transformer architecture for driver distraction detection. It leverages global self-attention across image patches while employing a pseudo-labeled multi-task self-training strategy that transfers emotional inductive cues from an independent teacher network without requiring manual emotion labels.

2. Core Architectural Concepts

Through my literature review, I have identified four critical components of the ViT-DD framework that guide my spatial baseline implementation:

- **Multi-Modal Visual Tokenization:**
    - The network receives two distinct spatial inputs: a global driver cabin view (224 × 224) and a localized cropped face image (32 × 32) extracted via RetinaFace.
    - Both images are partitioned into non-overlapping 16 × 16 patches (P=16). This produces 196 driver patches and 4 face patches (200 visual tokens total), which are linearly projected into D = 768 dimensional embeddings with added 1D learnable position embeddings.
    - Two task-specific tokens are prepended to the sequence: a distraction class token and an emotion class token, creating an aggregated input sequence of 202 tokens.
- **Global Self-Attention & Token Interaction:**
    - All 202 tokens pass through a shared 12-layer Transformer encoder (L=12, H=12 heads) based on a ViT-B backbone.
    - Using Multi-Head Self-Attention (MSA), the distraction token models pairwise correlations across all driver and face patches simultaneously.
    - Attention map analyses reveal that early layers capture broad cabin context, whereas deeper layers (Layers 7–12) concentrate sharp attention weights directly onto the steering wheel, handheld objects, and driver eye regions. Concurrently, the distraction and emotion class tokens begin attending heavily to each other in deeper layers to refine feature representations.
- **Pseudo-Labeled Multi-Task Self-Training:**
    - To circumvent the lack of emotion labels in driving datasets, a "Teacher" ViT is first trained on the external AffectNet-7 facial expression dataset.
    - The frozen Teacher infers pseudo emotion labels on face crops generated from the driver distraction dataset. If a face cannot be localized (e.g., driver turning completely away), a blank canvas is fed with a fallback Non-Face label.
    - The "Student" ViT-DD model is then optimized end-to-end via a weighted multi-task cross-entropy loss over both the ground-truth posture classes and the pseudo emotion classes.
- **Attention-Only Parameter Fine-Tuning:**
    - Because driving posture datasets are moderately sized, training the full Vision Transformer from scratch risks severe overfitting. The authors initialize the model with ImageNet pre-trained weights and fine-tune only the Multi-Head Self-Attention blocks, keeping the Multi-Layer Perceptron (MLP) layers frozen.

3. Relevance to CS-412 Project (2D vs. 3D Benchmark)

For our benchmark project (Real-Time Driver Distraction Detection: A 2D vs. 3D Vision Benchmark), ViT-DD provides the state-of-the-art Spatial / Vision Transformer Baseline to benchmark against both traditional 2D CNNs (Baheti et al.) and Huy's 3D spatio-temporal SlowFast network.

- **Strengths for DMS (Driver Monitoring System):**
    - **Superior Spatial Generalization:** On the challenging and realistic Split-by-Driver evaluation (testing strictly on unseen participants), ViT-DD achieves 93.59% accuracy on the AUC Distracted Driver Dataset (AUCDD), surpassing genetic CNN ensembles, ADNet, and hybrid Conv-LSTMs.
    - **Posture Disambiguation via Auxiliary Features:** Incorporating emotion cues provides strong regularizing inductive bias, boosting detection accuracy on ambiguous classes like Safe Driving (+13%) and Talking to Passenger (+9%) by linking neutral expressions to safe driving and smile/surprise to passenger interaction.
- **Challenges & Implementation Considerations for DMS:**
    - **Lack of Temporal Modeling:** Because ViT-DD operates on static 2D frames, it suffers from misclassifications during dynamic transitions (e.g., confusing Reaching Behind with Drinking or Adjusting Radio with Safe Driving) where temporal continuity is needed. This defines the exact performance gap we aim to measure against Huy's SlowFast model.
    - **Edge Latency & Asymmetric Occlusion:** ViT attention layers present a higher computational footprint than lightweight CNNs on edge hardware. Additionally, dataset camera positioning (mounted on the passenger side) introduces severe occlusion on left-hand interactions (Phone Left at 89% vs. Phone Right at 100%), highlighting the need to evaluate spatial models under varying real-world camera viewpoints.